

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
6	(a)	(i)		phytomining uses lower grade ores than bioleaching ☐ bioleaching conserves supplies of the lowest grade metal ores ☐ phytomining conserves supplies of high-grade metal ores ☒ bioleaching uses high-grade metal ores ☐ phytomining uses the lowest grade metal ores ☐			1	1		
		(ii)		yes – plants and bacteria are both living things / living organisms must give reason			1	1		
		(iii)		award (1) for any of following - insufficient land (available to grow plants) - land is used for other things e.g. food crops - unsuitable climate (to grow plants) e.g. not sunny enough / lack of water / too hot accept any sensible answer			1	1		
		(iv)		award (1) for either of following <u>bacteria</u> extract (the copper) at a low rate <u>bacterial</u> growth is slow / have to wait for the bacteria to grow			1	1		

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	(b)			iron is more reactive than copper (1) iron can displace copper / can reduce copper ions from the solution / (1) accept a correct equation for the second mark only e.g. $\text{Fe} + \text{CuSO}_4 \rightarrow \text{Cu} + \text{FeSO}_4$ neutral answers – reference to supply or cost		2		2		2
	(c)			ratio of Cu : O atoms $\Rightarrow \frac{2.54}{63.5} : \frac{0.32}{16}$ (1) 0.04 : 0.02 (1) 2:1 ratio $\Rightarrow \text{Cu}_2\text{O}$ (1) award (1) for correct answer with no working ecf possible award (2) for CuO_2 with working – incorrect division of mass and A_r		3		3	3	
				Question 6 total	0	5	4	9	3	2